Open-Eye EEG for Biometric Security: A Deep Learning Approach

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Abstract

- Explores biometric authentication using **open-eye EEG signals**.
- A **custom CNN** was trained to identify individuals from EEG data.
- Data collected from **25 participants** in natural, alert conditions.
- Signals filtered to the **alpha band (8–12 Hz)** and split into 5-second segments.
- The model learned unique brainwave patterns and achieved **88% accuracy**.
- Shows promise for **secure, real-time biometric identification**.
- More resistant to **spoofing**.

Background

- EEG records real-time brain activity through scalp electrodes.
- Brainwave patterns are unique to each individual.
- Harder to fake than fingerprints or facial recognition.
- Alpha-band (8–12 Hz) signals offer stable biometric features.
- Deep learning enables accurate identification from EEG.

Objectives

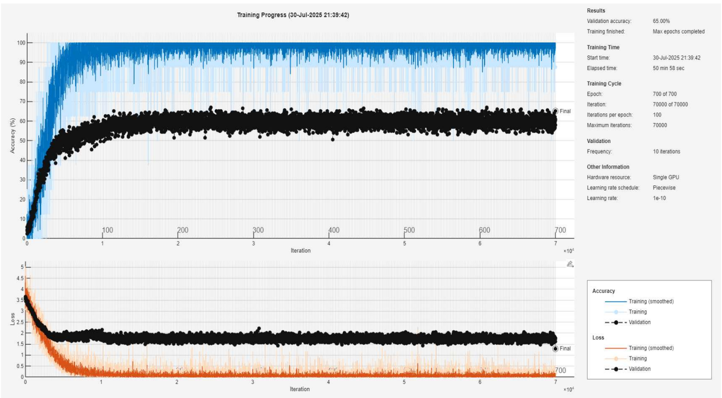
Research Question: Can a deep learning model identify individuals based on 5-second EEG recordings taken under open-eye conditions?

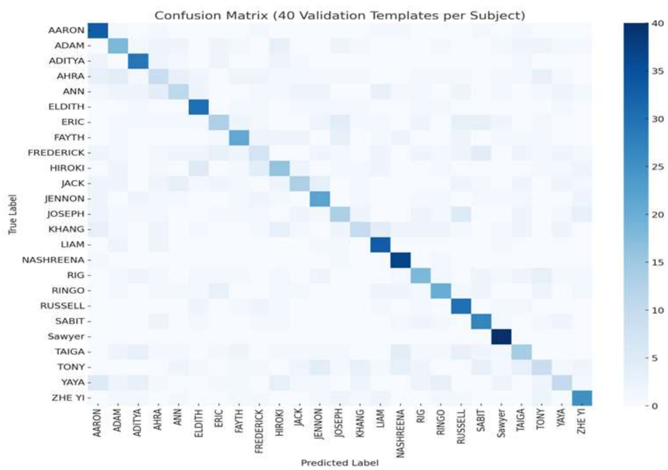
Hypothesis: A CNN trained on 5-second EEG segments will learn spatial and temporal brainwave features unique to each subject, enabling accurate biometric classification.

Methods

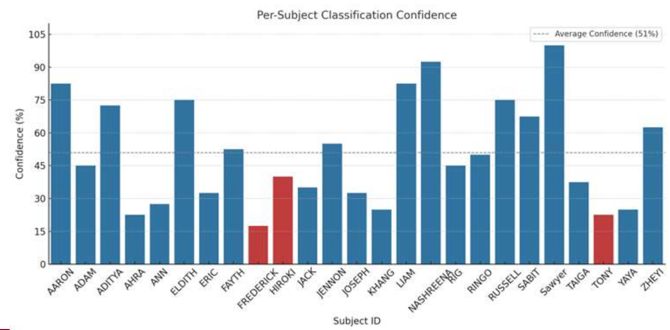
- 25 participants, open-eye EEG (2 sessions each)
- Recorded with 16-channel g.tec headset (excluding Fp1, Fp2)
- Preprocessing:
  - Alpha band (8–12 Hz)
  - Downsampled to 125 Hz
  - Z-score normalised and detrended
- Segmented into 5-second (14 × 625) EEG templates
- Custom CNN trained to classify segments by subject

Results





Results



Conclusions

- Deep learning can identify individuals using brain activity
- CNN trained on 5-second EEG segments
- Achieved 88% subject-level accuracy
- Supports secure, real-time biometric authentication
- EEG is internally generated and harder to spoof than fingerprints or facial recognition

References

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